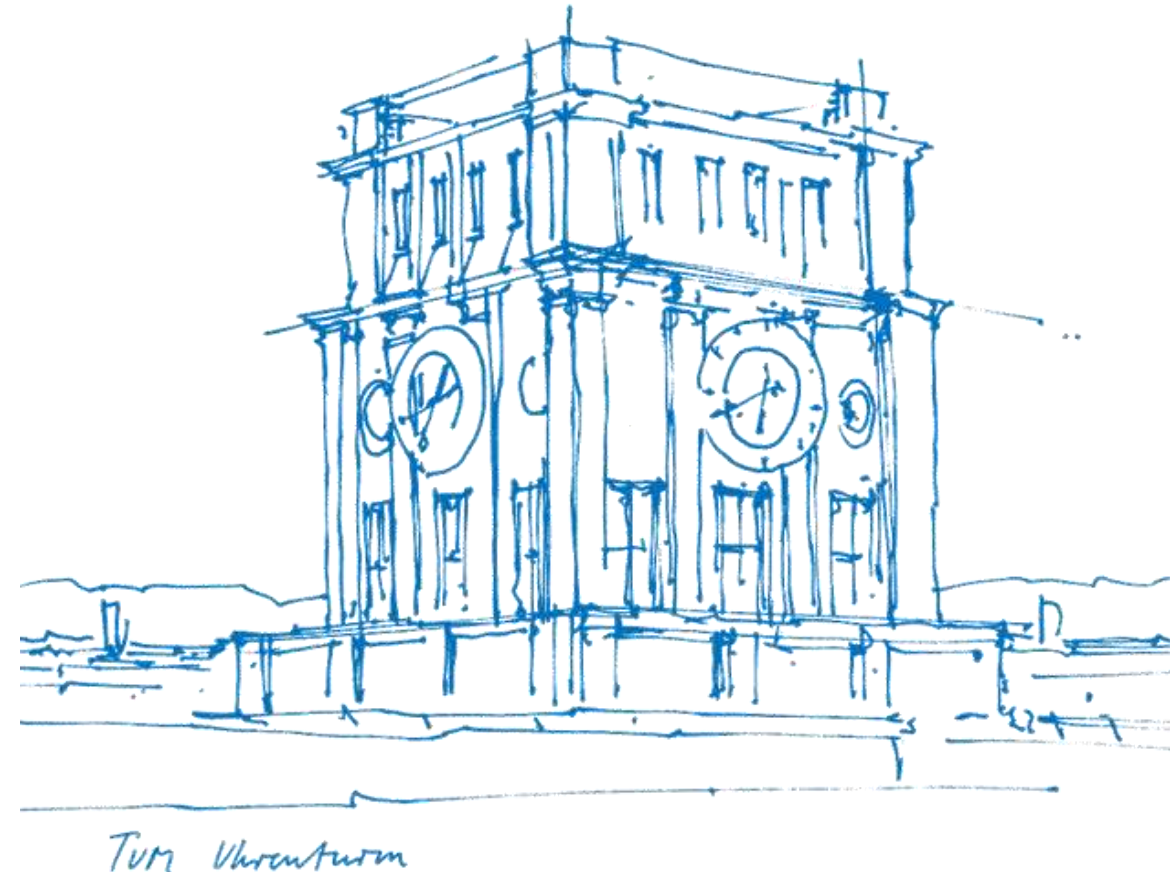


Tutorial 01

σ - Algebras

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About Tutorial

■ **Tutorial:** In-Person, 1.5h by two time periods per week - 12:15-13:45 Mon, 00.08.053 (A31)
- 14:15-15:45 Tue, 00.08.053 (B42)

■ The slides are not guaranteed to be accurate! (Written by myself)

■ **Materials of this tutorial:** Please scan the QR-Code 

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- **PART I.** Recap of the Lecture
- **PART II.** Exercises from the Exercise Sheet
- **PART III.** Additional Exercises (with solution)



I. Recap

- **σ - Algebra**

- Definition
- Intersection of σ -algebras
- Generator of a σ -algebra

II. Exercise Sheet

Exercise 1. Let $\Omega \neq \emptyset$ and let $\{A_\alpha \in \mathcal{P}(\Omega) \mid \alpha \in I\}$ be an arbitrary set of subsets of Ω , I the indexing set. Then the following two identities hold (commonly referred to as *de Morgan's law*):

$$\left(\bigcap_{\alpha} A_{\alpha} \right)^c = \bigcup_{\alpha} A_{\alpha}^c, \quad \left(\bigcup_{\alpha} A_{\alpha} \right)^c = \bigcap_{\alpha} A_{\alpha}^c$$

II. Exercise Sheet

Exercise 2. Let Ω_1, Ω_2 be non-empty sets, $f : \Omega_1 \rightarrow \Omega_2$ an arbitrary mapping and $\mathcal{C} \subseteq \mathcal{P}(\Omega_2)$ an arbitrary collection of subsets of Ω_2 . Then the following statements hold.

- (i) If $A, B \subseteq \Omega_2$ and $A \subseteq B$, then $f^{-1}(A) \subseteq f^{-1}(B)$.
- (ii) The inverse image of the union is equal to the union of the inverse images, meaning

$$f^{-1}\left(\bigcup_{A \in \mathcal{C}} A\right) = \bigcup_{A \in \mathcal{C}} f^{-1}(A) .$$

- (iii) Given a subset $A \subseteq \Omega_2$, the inverse-image of the complement is equal to the complement of the inverse image, meaning

$$f^{-1}(A^c) = (f^{-1}(A))^c$$

II. Exercise Sheet

Exercise 3. Suppose $\mathcal{A}$ is a σ -algebra over $\Omega \neq \emptyset$, let $A_0, A_1, A_2, \dots$ be a countable collection of sets in $\mathcal{A}$ and let $m \in \mathbb{N}$. Then the sets

$$\Omega, \quad A_0 \setminus A_1, \quad \bigcup_{i=0}^m A_i, \quad \bigcap_{i=0}^m A_i, \quad \bigcap_{i \in \mathbb{N}} A_i$$

are also in $\mathcal{A}$.

[Hint: Use de Morgan's law for the intersection properties.]

II. Exercise Sheet



Exercise 4. True or false? Justify your answers either by a proof or a counterexample.

- (i) The set $\mathcal{A} = \{\emptyset, \{1, 2\}, \{1, 4\}, \{2, 3\}, \{3, 4\}, \{1, 2, 3, 4\}\} \subseteq \mathcal{P}(\Omega)$ is a σ -algebra over $\Omega = \{1, 2, 3, 4\}$.
- (ii) If $\mathcal{A}_1, \mathcal{A}_2 \subseteq \mathcal{P}(\Omega)$ are σ -algebras over Ω , then so is $\mathcal{A}_1 \cup \mathcal{A}_2$.
- (iii) The set $\mathcal{A} := \{A \subseteq \Omega \mid A \text{ or } A^c \text{ is countable}\}$ is a σ -algebra over Ω .
- (iv) The set $\mathcal{A} := \{A \subseteq \Omega \mid A \text{ or } A^c \text{ is finite}\}$ is a σ -algebra over Ω if and only if Ω is finite.

[Hint: You can use the following facts without further justification: The countable union of countable sets is again countable. Subsets of countable sets are again countable.]

II. Exercise Sheet



Exercise 5. Let $\Omega \neq \emptyset$ be a set, I be a nonempty index set (finite, countable or even uncountable) and let $\mathcal{A}_\alpha$ be a σ -algebra over Ω for each $\alpha \in I$ in the index set. Then $\mathcal{A} := \bigcap_{\alpha \in I} \mathcal{A}_\alpha$, the intersection of all the sigma-algebras is also a σ -algebra over Ω .

II. Exercise Sheet

Exercise 6. Let Ω_1, Ω_2 be nonempty sets, $f : \Omega_1 \rightarrow \Omega_2$ an arbitrary mapping and let $\mathcal{A}, \mathcal{B}$ be σ -algebras over Ω_1 and Ω_2 respectively. Then the *inverse image of $\mathcal{B}$ under f* and the *image or push-forward of $\mathcal{A}$ under f* defined as

$$f^{-1}(\mathcal{B}) := \{f^{-1}(B) \mid B \in \mathcal{B}\} \quad \text{and} \quad f_*(\mathcal{A}) := \{B \subseteq \Omega_2 \mid f^{-1}(B) \in \mathcal{A}\}$$

are σ -algebras over Ω_1 and Ω_2 respectively.

[Hint: Use statements from the second exercise.]

III. Additional Exercises

Exercise 7. Anna flips a fair coin countable but infinitely many times. The set Ω consists of all infinite sequences of the outcomes "Heads" and "Tails".

(a) Show that the set Ω is uncountable.

(b) Let $\mathcal{A}$ be the set of all subsets $A \subseteq \Omega$ such that either A and $\Omega \setminus A$ is countable. Prove that $\mathcal{A}$ is a σ -Algebra.

(a) Assume every natural number i maps to a sequence of coins with i heads, the sequence with all tails is then not included, therefore Ω uncountable.

(b) $\mathcal{A} = \{A \subseteq \Omega \mid A \text{ or } \bar{A} \text{ countable}\}$.

Rule 1. $\Omega \in \mathcal{A}$, since $\Omega \subseteq \Omega$ and $\bar{\Omega} = \emptyset$ countable.

Rule 2. $A \in \mathcal{A}$, A or $\bar{A}$ countable $\Rightarrow \bar{A} \in \mathcal{A}$, since $\bar{A} \subseteq \Omega$.

Rule 3. ① if every A_n countable, $\bigcup_{n=1}^{\infty} A_n$ countable, $\bigcup_{n=1}^{\infty} A_n \in \mathcal{A}$.
② if exists an uncountable A_i , $\bar{A}_i = \Omega \setminus A_i$ must be countable, $\left(\bigcup_{n=1}^{\infty} A_n\right) = \Omega \setminus \bigcup_{n=1}^{\infty} \bar{A}_n \in \Omega \setminus A_i$ countable. $\Rightarrow$ For $\forall n, A_n \in \mathcal{A}$, then $\bigcup_{n=1}^{\infty} A_n \in \mathcal{A}$.

Materials

